

HR System — Database Structure

18 Tables · SQLAlchemy ORM · PostgreSQL Backend · Generated 2026-04-11

■ Primary Key row

■ Foreign Key row

All tables include created_at and updated_at timestamp columns (server default: now())

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Core HR

1. Employee

Table: employees

Central employee record. Every other table relates back to this one.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_number	String(20)	NOT NULL	UNIQUE, indexed — company ID (e.g. EMP001)
first_name	String(100)	NOT NULL	
last_name	String(100)	NOT NULL	
full_name	String(200)	NOT NULL	Stored for quick display
nic_number	String(20)	nullable	UNIQUE — Sri Lanka National ID
date_of_birth	Date	nullable	
gender	Enum	nullable	Values: MALE, FEMALE, OTHER
personal_email	String(200)	NOT NULL	UNIQUE
work_email	String(200)	nullable	UNIQUE
phone_number	String(100)	nullable	
address	Text	nullable	
city	String(100)	nullable	
district	String(100)	nullable	Sri Lanka district
hashed_password	String(255)	NOT NULL	Bcrypt hash for authentication
is_active	Boolean	NOT NULL	Default: True
last_login	DateTime	nullable	
department_id	Integer	nullable	FOREIGN KEY → departments.id
role_id	Integer	nullable	FOREIGN KEY → roles.id
manager_id	Integer	nullable	FOREIGN KEY → employees.id (self-reference)
hire_date	Date	nullable	
probation_end	Date	nullable	
termination_date	Date	nullable	

Field / Column	Data Type	Nullable	Constraints / Notes
employment_type	String(50)	NOT NULL	Default: full_time full_time / part_time / contract / intern
status	Enum	NOT NULL	Default: PROBATION ACTIVE, INACTIVE, ON_LEAVE, TERMINATED, PROBATION
base_salary	Numeric(12,2)	nullable	Monthly salary in LKR
bank_account	String(50)	nullable	
bank_name	String(100)	nullable	
face_embedding	JSON	nullable	DeepFace vector array (not the photo itself)
face_registered	Boolean	NOT NULL	Default: False
face_registered_at	Text	nullable	ISO datetime string
language_pref	String(5)	NOT NULL	Default: en en / si / ta
profile_photo	String(500)	nullable	Path to profile image
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

2. Department

Table: departments

Organisational units (Engineering, HR, Finance, ...).

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
name	String(100)	NOT NULL	UNIQUE — e.g. Engineering
code	String(20)	NOT NULL	UNIQUE — e.g. ENG
description	Text	nullable	
is_active	Boolean	NOT NULL	Default: True
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

3. Role

Table: roles

Job roles with access levels. access_level: 1=Employee, 2=HR Staff, 3=HR Manager, 4=Admin.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
title	String(100)	NOT NULL	UNIQUE — e.g. Software Engineer
code	String(30)	NOT NULL	UNIQUE — e.g. SWE
description	Text	nullable	
access_level	Integer	NOT NULL	Default: 1 1=Employee ... 4=Admin
base_salary	Numeric(12,2)	nullable	Base salary in LKR
is_active	Boolean	NOT NULL	Default: True
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

Attendance

4. Attendance

Table: attendance

Daily attendance record per employee. Includes face-recognition confidence, GPS, and overtime data.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
work_date	Date	NOT NULL	Indexed — the calendar date
clock_in	DateTime(tz)	nullable	Timezone-aware check-in timestamp
clock_out	DateTime(tz)	nullable	Timezone-aware check-out timestamp
work_hours	Float	nullable	Calculated hours worked (e.g. 8.5)
confidence_score	Float	nullable	DeepFace confidence 0.0–1.0
verification_method	Enum	NOT NULL	Default: FACE_RECOGNITION FACE_RECOGNITION, MANUAL_OVERRIDE, RFID
is_verified	Boolean	NOT NULL	Default: False
attendance_type	Enum	NOT NULL	Default: REGULAR REGULAR, OVERTIME, HALF_DAY, HOLIDAY
is_late	Boolean	NOT NULL	Default: False
late_minutes	Integer	NOT NULL	Default: 0
is_early_departure	Boolean	NOT NULL	Default: False
overtime_hours	Float	NOT NULL	Default: 0.0
location	String(200)	nullable	e.g. Head Office, Remote
latitude	Float	nullable	GPS latitude at clock-in
longitude	Float	nullable	GPS longitude at clock-in
checkout_latitude	Float	nullable	GPS latitude at clock-out
checkout_longitude	Float	nullable	GPS longitude at clock-out
is_absent	Boolean	NOT NULL	Default: False — no clock-in recorded
absence_reason	Text	nullable	
notes	Text	nullable	HR notes
flagged	Boolean	NOT NULL	Default: False — suspicious activity

Field / Column	Data Type	Nullable	Constraints / Notes
flag_reason	Text	nullable	e.g. Low confidence score
clock_in_photo	String(500)	nullable	Path to saved snapshot
clock_out_photo	String(500)	nullable	Path to saved snapshot
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

5. AttendanceScan

Table: attendance_scans

Individual scan events linked to an attendance record (supports multiple scans per day).

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
attendance_id	Integer	NOT NULL	FOREIGN KEY → attendance.id, indexed
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
scan_type	String(20)	NOT NULL	clock_in or clock_out
scanned_at	DateTime(tz)	NOT NULL	Exact scan timestamp (timezone-aware)
confidence	Float	nullable	Face recognition confidence
photo_path	String(500)	nullable	Path to saved snapshot
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

Leave Management

6. LeaveType

Table: leave_types

Catalogue of leave categories with eligibility rules.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
name	String(100)	NOT NULL	UNIQUE — e.g. Annual Leave, Sick Leave
code	String(20)	NOT NULL	UNIQUE — e.g. AL, SL, ML
description	Text	nullable	
max_days_per_year	Integer	NOT NULL	Total days allowed per year
max_consecutive_days	Integer	nullable	Max days in one request
requires_document	Boolean	NOT NULL	Default: False — e.g. medical cert required
is_paid	Boolean	NOT NULL	Default: True
gender_specific	String(10)	nullable	female for maternity, NULL for all
is_active	Boolean	NOT NULL	Default: True
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

7. LeaveBalance

Table: leave_balances

Per-employee, per-leave-type annual balance tracker.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
leave_type_id	Integer	NOT NULL	FOREIGN KEY → leave_types.id
year	Integer	NOT NULL	e.g. 2025, 2026
total_days	Float	NOT NULL	Days allocated this year
used_days	Float	NOT NULL	Default: 0.0 — days already taken
pending_days	Float	NOT NULL	Default: 0.0 — days in pending requests
remaining_days	Float	NOT NULL	total_days - used_days - pending_days
carried_over	Float	NOT NULL	Default: 0.0 — carried from previous year
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

8. LeaveRequest

Table: leave_requests

Employee leave applications with AI decision tracking and HR override capability.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
leave_type_id	Integer	NOT NULL	FOREIGN KEY → leave_types.id
start_date	Date	NOT NULL	
end_date	Date	NOT NULL	
total_days	Float	NOT NULL	Working days between start and end
days_requested	Float	nullable	Alias for total_days
reason	Text	NOT NULL	Employee's stated reason
is_half_day	Boolean	NOT NULL	Default: False
status	Enum	NOT NULL	Default: PENDING, indexed PENDING, APPROVED, REJECTED, ESCALATED, CANCELLED, ON_LEAVE, COMPLETED
ai_decision	String(20)	nullable	approved or rejected
ai_decision_reason	Text	nullable	LLM explanation
ai_confidence	Float	nullable	AI confidence 0.0–1.0
ai_policy_refs	JSON	nullable	Policy chunks used: [Section 4.2, ...]
ai_processed_at	Text	nullable	ISO datetime when AI processed
rejection_reason	Text	nullable	Reason for rejection
approved_by	String(200)	nullable	AI Agent or HR employee name
approved_at	DateTime	nullable	When approved / rejected
reviewed_by_id	Integer	nullable	FOREIGN KEY → employees.id (HR reviewer)
hr_override	Boolean	NOT NULL	Default: False — HR overrode AI decision
hr_notes	Text	nullable	
reviewed_at	Text	nullable	ISO datetime
document_url	String(500)	nullable	S3 URL of supporting document
document_verified	Boolean	NOT NULL	Default: False

Field / Column	Data Type	Nullable	Constraints / Notes
is_appealed	Boolean	NOT NULL	Default: False
appeal_reason	Text	nullable	
appeal_status	String(20)	nullable	pending, upheld, or overturned
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

Performance

9. PerformanceReview

Table: performance_reviews

Period-based employee performance evaluations (attendance, punctuality, OT). Supports AI summaries.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
reviewer_id	Integer	nullable	FOREIGN KEY → employees.id (null if AI-generated)
period_type	Enum	NOT NULL	MONTHLY, QUARTERLY, ANNUAL, PROBATION
period_start	Date	NOT NULL	
period_end	Date	NOT NULL	
attendance_score	Float	nullable	$(\text{present_days} / \text{working_days}) \times 100$
punctuality_score	Float	nullable	$100 - (\text{late_days} / \text{present_days} \times 100)$
overtime_score	Float	nullable	Based on voluntary overtime
overall_score	Float	nullable	Weighted average — used by AI
rating	String(50)	nullable	Excellent, Good, Satisfactory, Needs Improvement
strengths	Text	nullable	
areas_to_improve	Text	nullable	
goals_next_period	Text	nullable	
manager_comments	Text	nullable	
ai_summary	Text	nullable	AI-generated narrative
status	Enum	NOT NULL	Default: DRAFT DRAFT, PENDING, COMPLETED, DISPUTED
is_promotion_eligible	Boolean	NOT NULL	Default: False
requires_pip	Boolean	NOT NULL	Default: False — Performance Improvement Plan
employee_acknowledged	Boolean	NOT NULL	Default: False
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

10. PerformanceMetric

Table: performance_metrics

Granular metric values attached to a PerformanceReview.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
review_id	Integer	NOT NULL	FOREIGN KEY → performance_reviews.id, indexed
metric_name	String(200)	NOT NULL	e.g. Days Present, Times Late
value	Float	NOT NULL	Raw number
score	Float	nullable	0–100 score derived from value
weight	Float	NOT NULL	Default: 1.0 — contribution to overall_score
note	Text	nullable	

Recruitment

11. JobPosting

Table: job_postings

Open job positions published by HR. Includes AI interview questions for the screening pipeline.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
title	String(200)	NOT NULL	Job title
department_id	Integer	nullable	FOREIGN KEY → departments.id
description	Text	NOT NULL	Full job description
requirements	Text	NOT NULL	Skills, experience, qualifications
responsibilities	Text	nullable	
salary_min	Numeric(12,2)	nullable	
salary_max	Numeric(12,2)	nullable	
employment_type	String(50)	NOT NULL	Default: full_time
location	String(200)	nullable	
positions_count	Integer	NOT NULL	Default: 1 — number of openings
closing_date	Date	nullable	
is_active	Boolean	NOT NULL	Default: True
posted_by_id	Integer	nullable	FOREIGN KEY → employees.id
ai_interview_questions	JSON	nullable	Questions for AI to ask candidates
culture_keywords	JSON	nullable	Keywords for RAG culture-fit matching
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

12. JobApplication

Table: job_applications

Candidate applications with AI screening scores (resume match, interview, culture fit).

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
job_posting_id	Integer	NOT NULL	FOREIGN KEY → job_postings.id, indexed
applicant_id	Integer	nullable	FOREIGN KEY → employees.id (internal applicants)
applicant_name	String(200)	nullable	External applicant name
applicant_email	String(200)	nullable	
applicant_phone	String(20)	nullable	
resume_url	String(500)	nullable	S3 URL of resume
cover_letter	Text	nullable	
status	Enum	NOT NULL	Default: APPLIED APPLIED, SCREENING, AI_INTERVIEW, SHORTLISTED, HR_INTERVIEW, OFFERED, ACCEPTED, REJECTED, WITHDRAWN
ai_resume_score	Float	nullable	0–100 resume match score
ai_interview_score	Float	nullable	0–100 from AI interview
ai_culture_fit	Float	nullable	0–100 culture match
ai_overall_score	Float	nullable	Weighted average of AI scores
ai_recommendation	String(50)	nullable	shortlist, reject, or hold
ai_feedback	Text	nullable	Detailed AI feedback
interview_transcript	JSON	nullable	Q&A; list: [{q, a, score}, ...]
hr_notes	Text	nullable	
reviewed_by_id	Integer	nullable	FOREIGN KEY → employees.id (HR reviewer)
hr_score	Float	nullable	
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

Communication

13. Notification

Table: notifications

In-app and email notifications with read-state tracking.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
notification_type	String(50)	NOT NULL	Indexed — e.g. leave_approved, payslip_ready
title	String(300)	NOT NULL	Short heading
message	Text	NOT NULL	Full message body
action_url	String(500)	nullable	Deep link e.g. /leave/request/42
channel	Enum	NOT NULL	Default: BOTH IN_APP, EMAIL, BOTH
is_read	Boolean	NOT NULL	Default: False
read_at	Text	nullable	ISO datetime
email_sent	Boolean	NOT NULL	Default: False
email_sent_at	Text	nullable	ISO datetime
related_entity_type	String(50)	nullable	leave_request, payroll, etc.
related_entity_id	Integer	nullable	ID of related record
priority	String(20)	NOT NULL	Default: normal low, normal, high, urgent
extra_data	JSON	nullable	Additional structured data
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

14. ChatSession

Table: chat_sessions

AI chat conversations. Title is auto-set from the first message.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	NOT NULL	FOREIGN KEY → employees.id, indexed
title	String(200)	nullable	Auto-set from first message
is_active	Boolean	NOT NULL	Default: True
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

15. ChatMessage

Table: chat_messages

Individual messages within a ChatSession.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
session_id	Integer	NOT NULL	FOREIGN KEY → chat_sessions.id, indexed
role	String(20)	NOT NULL	user or assistant
content	Text	NOT NULL	Message text
sources	Text	nullable	JSON list of policy sources used by AI
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

Governance

16. AuditLog

Table: audit_logs

Immutable log of every action taken by employees, HR, admins, and AI agents.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
employee_id	Integer	nullable	FOREIGN KEY → employees.id, indexed (null for system/AI)
actor_type	String(20)	NOT NULL	employee, hr_manager, admin, ai_agent, system
ip_address	String(50)	nullable	
action	String(100)	NOT NULL	Indexed — dot notation e.g. leave.approved, employee.login
description	Text	nullable	Human-readable e.g. AI approved leave #42 for John
entity_type	String(50)	nullable	leave_request, employee, payroll, etc.
entity_id	Integer	nullable	ID of affected record
before_state	JSON	nullable	State BEFORE action (for undo)
after_state	JSON	nullable	State AFTER action
metadata	JSON	nullable	Extra info: AI reasoning, policy refs, etc.
status	String(20)	NOT NULL	Default: success success, failed, blocked
error_message	Text	nullable	
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

17. HRPolicy

Table: hr_policies

Policy documents stored in PostgreSQL and chunked into ChromaDB for RAG retrieval.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
title	String(300)	NOT NULL	e.g. Annual Leave Policy 2025
category	Enum	NOT NULL	Indexed LEAVE, ATTENDANCE, PAYROLL, CONDUCT, RECRUITMENT, PERFORMANCE, GENERAL
content	Text	NOT NULL	Full policy text (chunked for ChromaDB)
version	String(20)	nullable	e.g. v2.1
effective_date	String(20)	nullable	ISO date e.g. 2025-01-01
expiry_date	String(20)	nullable	ISO date
chroma_doc_id	String(200)	nullable	UNIQUE — links PostgreSQL row to ChromaDB document
is_indexed	Boolean	NOT NULL	Default: False — already in ChromaDB
indexed_at	Text	nullable	ISO datetime
chunk_count	Integer	nullable	Number of chunks created in ChromaDB
is_active	Boolean	NOT NULL	Default: True
uploaded_by_id	Integer	nullable	Employee ID of uploader
tags	JSON	nullable	Filter keywords e.g. [annual, leave, eligibility]
language	String(5)	NOT NULL	Default: en en, si, ta
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()

18. HRReport

Table: hr_reports

Generated analytics reports (monthly, quarterly, annual) with AI executive summaries.

Field / Column	Data Type	Nullable	Constraints / Notes
id	Integer	NOT NULL	PRIMARY KEY, auto-increment
report_type	String(100)	NOT NULL	Indexed — monthly_summary, attendance, leave, performance, department
period	String(50)	nullable	e.g. 2026-03, 2026-Q1, 2026
title	String(300)	NOT NULL	
content	JSON	NOT NULL	Default: {} — structured KPIs and data
narrative	Text	nullable	AI-generated executive summary
generated_by	String(200)	NOT NULL	Default: system AI Agent or employee name
created_at	DateTime	NOT NULL	Server default: now()
updated_at	DateTime	NOT NULL	Server default: now(), onupdate: now()